



Diopside

△ Fine diopside crystals on a rock groundmass

Bright green diopside would make a superb gemstone to rival emerald in colour, were it not for its softness and easily set-off cleavage. It is very popular with mineral collectors, but its extreme fragility means that, although it can be cut into a collector's gemstone, these are very susceptible to mechanical shock, and will shatter if exposed to ultrasonic cleaning; even mineral specimens of diopside must be carefully handled and stored. The name diopside refers to the gem's highly transparent crystals, from the Greek for "to see through".

Specification

Chemical name	Copper silicate	Formula	$\text{CuSiO}_2(\text{OH}_2)$
Colours	Emerald- to blue-green	Structure	Hexagonal/trigonal
Hardness	5	SG	3.3
RI	1.67–1.72	Lustre	Vitreous to greasy
Streak	Pale greenish-blue	Locations	Kazakhstan, Iran, Namibia, Congo, Argentina, Chile, USA



Diopside crystals

Diopside on quartz | Rough | This spectacular specimen of diopside crystals on quartz shows why it is a favourite with mineral collectors as well as gemologists.



Large crystals on quartz | Rough | In this specimen, diopside crystals of remarkable form are highlighted by overgrowths of the mineral plancheite.



Gemstone with inclusions | Cut | Although this diopside gem has a large number of internal inclusions, its skilled cut and rich colour still make it a fine gem.

Diopside crystals, mistakenly identified as emeralds, were given to Tsar Paul I in 1797



Diamonds

Gold mount

Brooch/pendant | Set | This piece can be worn as a brooch or a pendant. It features 14-karat gold mounting a natural cluster of diopside highlighted with diamonds.



Superb crystals | Colour variety | This striking group of diopside crystals displays unusually fine crystallization and a deep blue-green colour.



Bird's nest pendant | Set | This intricately textured gold pendant in the shape of a nest has a centrally mounted group of natural diopside crystals.